# Light Bulb & Resistor

Light bulbs do not have a linear current vs. voltage characteristic. As a consequence, the simple rule for resistors in series does not hold true anymore. In this lab you will learn how to analyse a dc circuit containing a light bulb and a resistor in series. An additional goal of this lab is to practice the graphical representation and analysis of data, in particular by means of curve fits..

## Measurements

### Characteristic of Light Bulb

Circuit Diagram:

Measure the current through a light bulb for different voltages across the light bulb.

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $V$ (V) | 23.97 | 22.02 | 20.05 | 18.01 | 16.03 | 14.01 | 12.02 | 10.01 | 8.00 | 6.03 | 4.010 | 2.002 | ± 0.03 |
| $I$ (mA) | 120.7 | 115.5 | 109.5 | 102.8 | 96.1 | 88.9 | 80.6 | 72.8 | 63.7 | 54.0 | 41.3 | 27.6 | ±0.5 |

### Analysis

a) Calculate the the resistance values (ratio of voltage and current) and graph the resistance vs current. Determine the parameters for a linear fit and their uncertainties (with correct units).

b) Derive the expected model for the current as a function of the voltage using the linear relation of a).

c) Graph the current vs. voltage and fit the expected function to the data. Determine the fit parameter and its uncertainty (with correct units). You may want to try other fit functions. Compare the results of a) and b).

### Series Circuit

Circuit Diagram:

Measure the partial voltages $V_R$ (across the resistor) and $V_L$ (across the light bulb) and the current $I$ in a circuit with the light bulb and a resistor connected in series to a fixed voltage $V_0$. Measure the resistance values for the resistors with a multimeter.$V_0$ =

| | | | | | |
|---|---|---|---|---|---|
| $R$ (Ω) | | | | | |
| $V_R$ (V) | | | | | |
| $V_L$ (V) | | | | | |
| $I$ (A) | | | | | |

#### Analysis

a) Make a diagram with the characteristic of the light bulb and the *load line* (i.e. the line representing the current through the resistor as a function of the voltage across the light bulb) for one of the resistors, and explain why the intersection of the two graphs allows you to determine the partial voltages for the series circuit.

b) For each of the resistors, determine the expected values for the partial voltages and the current using the load line and the fit function for the light bulb. Compare the calculated to the measured values.

## Additional Tasks

1. Predict the partial voltages and currents in a more complicated circuit containing resistors and (identical) light bulbs, e.g. a first light bulb in series to a light bulb and a resistor in parallel. The analysis has to be based on the characteristic of the light bulb and Ohm's law.
2. Find an explanation why the resistance of the filament in a light bulb is proportional to the current.
3. Simulate the discharging of a capacitor through a light bulb. To do this, set up a differential equation similar to the one for an RC circuit, but with a term for the current through the light bulb based on the square root fit. Solve the equation numerically and compare the solution to the solution for an RC circuit with a constant resistance.
4. Curve fits are usually based on the method of least square deviations. Find out how this method works for a simple function (e.g. a proportionality). Derive an expression for the fit parameter of a square root fit. Implement your result as a Python program, and apply it to the light bulb data.